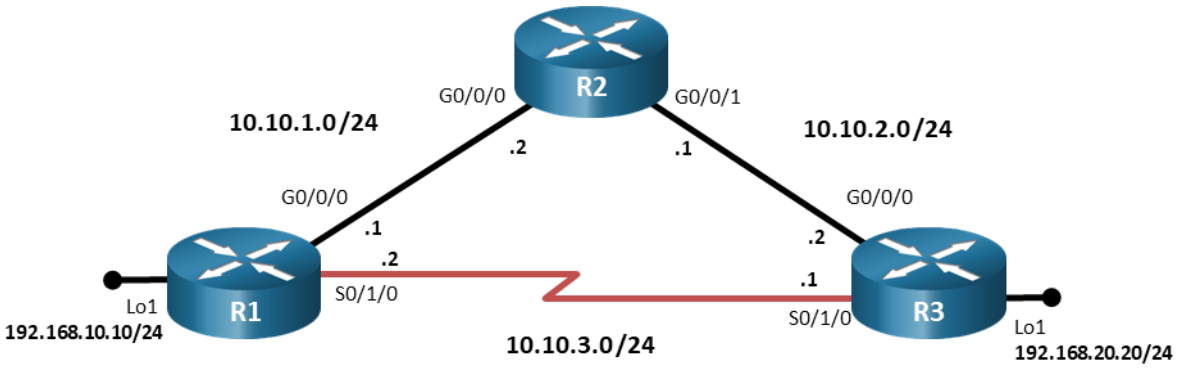
Lab - Troubleshoot uRPF

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask
R1	G0/0/0	10.10.1.1	255.255.255.0
	S0/1/0	10.10.3.2	255.255.255.0
	Lo1	192.168.10.10	255.255.255.0
R2	G0/0/0	10.10.1.2	255.255.255.0
	G0/0/1	10.10.2.1	255.255.255.0
R3	G0/0/0	10.10.2.2	255.255.255.0
	S0/1/0	10.10.3.1	255.255.255.0
	Lo1	192.168.20.20	255.255.255.0

Objectives

Troubleshoot issues related to the configuration and operation of uRPF.

Background / Scenario

At your company, an internal security review identified an increased risk of spoofed-source traffic traversing the WAN edge. To strengthen the network's security posture without impacting legitimate operations, the network team is rolling out unicast Reverse Path Forwarding (uRPF) on selected interfaces in a staged manner. This lab mirrors a small enterprise environment with three routers, EIGRP-based routing, and loopback interfaces representing internal services hosted on distinct subnets. Your objective is to validate that the uRPF rollout preserves expected reachability between all networks while maintaining the intended security controls.

In this lab, you will be loading configurations with intentional errors onto the network. Your tasks are to FIND the error(s), document your findings and the command(s) or method(s) used to fix them, FIX the issue(s) presented here, and then test the network to ensure both of the following conditions are met:

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1) The trouble ticket has been resolved
2) The network is fully functioning

Note: The routers used with CCNP hands-on labs are Cisco 8200L with Cisco IOS XE Release 17.12.05c (universalk9 image). Other routers and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs. Refer to the Router Interface Summary Table at the end of the lab for the correct interface identifiers.

Note: Make sure that the devices have been erased and have no startup configurations. If you are unsure, contact your instructor.

Required Resources

- 3 Routers (Cisco 8200L with Cisco IOS XE Release 17.12.05c universal image or comparable)
- Console cables to configure the Cisco IOS devices via the console ports
- 1 PC (Choice of operating system with a terminal emulation program installed)
- Ethernet and serial cables as shown in the topology

Instructions

Part 1: Trouble Ticket 32.1.3.1

Scenario:

During an approved maintenance window, uRPF was applied on R1 as part of the phased security hardening plan. Shortly after the change, operations began receiving reports that systems within the 192.168.20.0/24 segment could no longer reach resources in the 192.168.10.0/24 segment. The change advisory board has asked the network team to investigate and restore full connectivity while keeping the new security control in place. You have console access to R1, R2, and R3. Validate the issue, identify the misconfiguration causing the reachability problem, correct it, and confirm that all routed interfaces and loopback networks can communicate as expected. Document your findings and the steps you took to resolve the incident.

Step 1: Cable the network as shown in the topology.

a. Attach the devices as shown in the topology diagram, and cable as necessary.

b. Use the commands listed below to load the configuration files for this trouble ticket:

Device	Command
R1	`copy flash:/enarsi/32.1.3.1-r1-config.txt running-config`
R2	`copy flash:/enarsi/32.1.3.1-r2-config.txt running-config`
R3	`copy flash:/enarsi/32.1.3.1-r3-config.txt running-config`

Note: Passwords on all devices are **cisco12345**.

Step 2: Troubleshoot Ticket.

Troubleshoot and repair the issue. All devices, including loopback addresses, should be able to ping each other.

Step 3: Complete the Ticket.

a. After you have fixed the ticket, change the MOTD on Router R1 using the following command:

banner motd # This is $(*hostname*) FIXED from ticket <*ticket number*> #

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b. Verify that uRPF is enabled, configured correctly and all devices, including loopback addresses, can ping each other. Then save the configuration by issuing the **wri** command.

c. Inform your instructor that you are have completed the ticket.

d. After the instructor approves your solution for this ticket, issue the **reset.now** privileged EXEC command on each device. This script will clear your configurations and reload the devices.

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